SYDE 671 / ASSIGNMENT 3 / FALL 2026

Photo mosaics. *Manual to automatic.*

Recovering a shared perspective from overlapping photographs.

Moksh Trehan · November 1, 2026



Figure 1. City panorama: automatic mosaic from 3 photographs. The source images, feature matches, and manual comparison are documented below.

3 scenes
10 selected mosaic photographs

2 rectifications
Nearest neighbor and bilinear, implemented from scratch

6 mosaics
3 manual and 3 automatic estimates

Part A estimates image alignment from manually chosen landmarks. Part B finds the correspondences automatically using Harris corners, adaptive non-maximal suppression, normalized patch descriptors, and four-point RANSAC. Both paths share the homography solver and inverse warp. The blending comparisons show how seam placement affects the final image separately from alignment accuracy.

Part A · Geometry

A.1 Source photographs

A.2 Recovering homographies

A.3 Warping and rectification

A.4 Manual mosaics

Part B · Correspondence

B.1 Harris corners and ANMS

B.2 Feature descriptors

B.3 Matching features

B.4 RANSAC and automatic mosaics

A.1

Source photographs

All photographs in this report were supplied by the author for this assignment. Three overlapping sets depict a basketball court, an indoor lounge, and a city view; two additional photographs show book covers for rectification. The intended mosaic model is camera rotation about a fixed center. Camera motion was not measured, and depth-dependent residuals are consistent with foreground parallax and departures from that model.

All coordinates use **(x, y)** pixel centers, with the origin at the top-left. Original files are preserved. The working images are antialiased and reduced to at most **1,400 pixels** on their longest side. Saved original-to-working matrices carry annotations into the resized image frame.

1. Basketball court

Three central frames of the basketball-court sweep (IMG_1242–1244) share the window wall and backboard. IMG_1243 is the reference.



Figure 2. IMG_1242. Working image: 1400 × 1050 pixels.



Figure 3. IMG_1243 · reference view. Working image: 1400 × 1050 pixels.



Figure 4. IMG_1244. Working image: 1400 × 1050 pixels.

The central three frames give a finite projective mosaic. The full six-image sweep crosses the projective horizon in this reference frame; IMG_1241, IMG_1245, and IMG_1246 are retained locally as supplemental inputs.

2. Lounge

Four overlapping lounge views, ordered IMG_1247, IMG_1248, IMG_1250, IMG_1249. IMG_1248 is the reference; the last two images are ordered by scene overlap rather than filename.



Figure 5. IMG_1247. Working image: 1400 × 1050 pixels.



Figure 6. IMG_1248 · reference view. Working image: 1400 × 1050 pixels.



Figure 7. IMG_1250. Working image: 1400 × 1050 pixels.



Figure 8. IMG_1249. Working image: 1400 × 1050 pixels.

3. City panorama

Three overlapping outdoor views (IMG_1251–1253), with IMG_1252 as the reference. Buildings provide static landmarks distributed across the lower half of the images; the sky is not used for manual annotations.



Figure 9. IMG_1251. Working image: 1400 × 1050 pixels.



Figure 10. IMG_1252 · reference view. Working image: 1400 × 1050 pixels.



Figure 11. IMG_1253. Working image: 1400 × 1050 pixels.

> **What the projective model assumes**
>
> A single homography describes a plane, or an arbitrary scene viewed by a camera rotating about a fixed center of projection. Handheld translation, changing depth, moving objects, and lens distortion can leave residual misalignment even when the fitted points agree. The source sets are retained so these limits can be inspected directly.

A.2

Recovering homographies

A homography maps a source point to a destination point up to scale: **[u, v, 1]ᵀ ~ H [x, y, 1]ᵀ**. Fixing $h_{33} = 1$ leaves eight unknowns. Each correspondence contributes two equations, so four nondegenerate pairs are the minimum. The manual mosaics use more than four pairs and solve an overdetermined system.

One correspondence

```
[ x  y  1  0  0  0  −ux  −uy ] h = u
[ 0  0  0  x  y  1  −vx  −vy ] h = v

h = [h11 h12 h13 h21 h22 h23 h31 h32]ᵀ
```

Stacking n correspondences gives $A \in \mathbb{R}^{2n\times 8}$ and $b \in \mathbb{R}^{2n}$. `computeH` centers each point set and scales its mean distance from the origin to $\sqrt{2}$, solves with `numpy.linalg.lstsq`, and transforms the solution back into pixel coordinates. Rank-deficient point sets are rejected. Normalization improves conditioning without changing the geometric model.

Denormalize

```
H = Tdestination⁻¹ Hnormalized Tsource; then H ← H / h33
```

The displayed matrices map the left source image to the right target image in **working-image coordinates**. They are rounded for readability; full-precision values and the annotated points are included in the measurement download. Errors below are residuals on the same landmarks used for fitting, not an independent estimate of photographic accuracy.

Scene	Source → target	Landmarks	Median residual (px)	Max residual (px)
Basketball court	IMG_1242 → IMG_1243	12	1.026	1.733
Basketball court	IMG_1243 → IMG_1244	12	1.093	2.155
Lounge	IMG_1247 → IMG_1248	9	1.187	2.647
Lounge	IMG_1248 → IMG_1250	10	0.852	1.807
Lounge	IMG_1250 → IMG_1249	10	0.662	2.420
City panorama	IMG_1251 → IMG_1252	10	0.688	1.048
City panorama	IMG_1252 → IMG_1253	11	0.393	0.717

A.2

Landmarks · Basketball court

IMG_1242 → IMG_1243

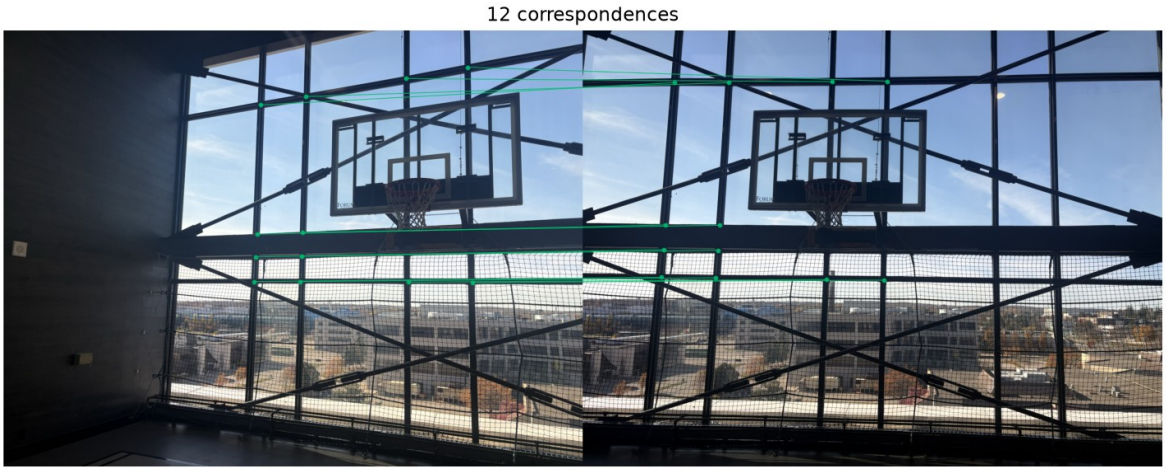


Figure 12. Basketball court: 12 manually selected corresponding landmarks, shown in green.

H · IMG_1242 → IMG_1243

$$\begin{bmatrix} 1.73134 & -0.0447851 & -771.757 \\ 0.323571 & 1.47209 & -298.927 \\ 0.000518232 & 1.9696e{-}05 & 1 \end{bmatrix}$$

IMG_1243 → IMG_1244



Figure 13. Basketball court: 12 manually selected corresponding landmarks, shown in green.

H · IMG_1243 → IMG_1244

$$\begin{bmatrix} 1.78429 & -0.0573305 & -853.753 \\ 0.345303 & 1.4712 & -288.963 \\ 0.000561144 & -3.36899e{-}05 & 1 \end{bmatrix}$$

A.2

Landmarks · Lounge

IMG_1247 → IMG_1248

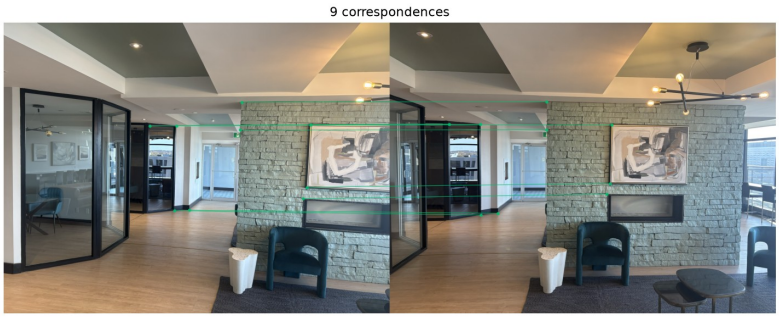


Figure 14. Lounge: 9 manually selected corresponding landmarks, shown in green.

H · IMG_1247 → IMG_1248

$$\begin{bmatrix} 1.50225 & 0.0633007 & -560.149 \\ 0.147034 & 1.33145 & -133.14 \\ 0.000361529 & 2.56486e{-}05 & 1 \end{bmatrix}$$

IMG_1248 → IMG_1250



Figure 15. Lounge: 10 manually selected corresponding landmarks, shown in green.

H · IMG_1248 → IMG_1250

$$\begin{bmatrix} 1.36997 & 0.0405726 & -459.218 \\ 0.124665 & 1.24884 & -116.916 \\ 0.000256585 & 3.0125e{-}05 & 1 \end{bmatrix}$$

IMG_1250 → IMG_1249



Figure 16. Lounge: 10 manually selected corresponding landmarks, shown in green.

H · IMG_1250 → IMG_1249

$$\begin{bmatrix} 1.28246 & -0.00828381 & -283.015 \\ 0.0951682 & 1.1299 & -61.084 \\ 0.00020838 & -3.29868e{-}05 & 1 \end{bmatrix}$$

A.2

Landmarks · City panorama

IMG_1251 → IMG_1252



Figure 17. City panorama: 10 manually selected corresponding landmarks, shown in green.

H · IMG_1251 → IMG_1252

$$\begin{bmatrix} 1.6121 & -0.0196081 & -610.387 \\ 0.245053 & 1.3938 & -232.292 \\ 0.000420283 & 3.57008e{-}05 & 1 \end{bmatrix}$$

IMG_1252 → IMG_1253



Figure 18. City panorama: 11 manually selected corresponding landmarks, shown in green.

H · IMG_1252 → IMG_1253

$$\begin{bmatrix} 1.85844 & -0.0382182 & -942.137 \\ 0.331829 & 1.51269 & -259.067 \\ 0.000635409 & -6.60971e{-}05 & 1 \end{bmatrix}$$

A.3

Inverse warping and interpolation

Forward splatting can leave holes. Instead, each output pixel is mapped backward by H^{-1} and sampled from the source. Automatic canvas bounds come from the transformed image corners. Rectification uses an explicit finite output rectangle. A separate validity mask distinguishes missing samples from valid black pixels.

Inverse mapping $[x', y', w']^\top = H^{-1}\,[u, v, 1]^\top;\quad (x, y) = (x'/w', y'/w')$

Nearest neighbor

Round the source coordinates to the nearest integer pixel and copy that value. The method keeps discrete source values but can produce jagged edges and blocky fine detail.

Sample

$$I_{out}(u,v) = I(\mathrm{round}(x), \mathrm{round}(y))$$

Bilinear interpolation

Let $x_0 = \mathrm{floor}(x)$, $y_0 = \mathrm{floor}(y)$, $a = x - x_0$, and $b = y - y_0$. Blend the four neighboring source pixels using the weights below. This smooths edge transitions at additional arithmetic cost.

Sample

$$(1-a)(1-b)\,I(x_0,y_0) + a(1-b)\,I(x_0+1,y_0) + (1-a)b\,I(x_0,y_0+1) + ab\,I(x_0+1,y_0+1)$$

Both methods are implemented directly in NumPy. They use the same closed domain of valid source pixel centers; the last row or column repeats its boundary neighbor for bilinear sampling. Neither method calls a library interpolation or projective-warp routine.

A.3 / 01

Letting Go front cover



Figure 19. Letting Go front cover: original with annotated corners, nearest-neighbor rectification, and bilinear rectification.



Figure 20. Identical title-region crops enlarged for inspection. Nearest-neighbor sampling shows sharper steps and broken fine strokes; bilinear sampling makes the edges smoother.

The four front-cover corners are mapped to (0, 0), (599, 0), (599, 899), and (0, 899). The output uses an approximate 2:3 width-to-height ratio because physical dimensions were not available. Straightening the quadrilateral therefore demonstrates projective rectification; it does not establish the object's true metric proportions. Physical cover curvature can remain.

H · working image → rectified image

$$\begin{bmatrix} 1.83648 & 0.20178 & -658.276 \\ 0.0188669 & 2.47747 & -1095.02 \\ -6.53106e{-}05 & 0.001049 & 1 \end{bmatrix}$$

A.3 / 02

The Magic of Thinking Big front cover



Figure 21. The Magic of Thinking Big front cover: original with annotated corners, nearest-neighbor rectification, and bilinear rectification.



Figure 22. Identical title-region crops enlarged for inspection. Nearest-neighbor sampling shows sharper steps and broken fine strokes; bilinear sampling makes the edges smoother.

The four front-cover corners are mapped to (0, 0), (599, 0), (599, 899), and (0, 899). The output uses an approximate 2:3 width-to-height ratio because physical dimensions were not available. Straightening the quadrilateral therefore demonstrates projective rectification; it does not establish the object's true metric proportions. Physical cover curvature can remain.

$$\text{H} \cdot \text{working image} \rightarrow \text{rectified image} = \begin{bmatrix} 0.82272 & 0.00169782 & -98.0427 \\ -0.109137 & 0.911069 & -94.3249 \\ -0.000251619 & 0.000140062 & 1 \end{bmatrix}$$

A.3

Measured speed and visible quality

Rectification	Output pixels	Nearest median (ms)	Bilinear median (ms)	Time ratio
Letting Go front cover	600 × 900	64.4	104.6	1.62×
The Magic of Thinking Big front cover	600 × 900	65.1	110.4	1.70×

One warmup per method; 9 timed runs each; alternate method order; warping only, excludes loading, H fitting and saving.

Environment: Python 3.12.14; macOS-15.5-arm64-arm-64bit. All timing samples are preserved in the rectification record.

Bilinear interpolation is slower in these measurements because each output sample combines four neighboring values. It produces more continuous letter contours in the close-ups, while nearest neighbor retains sharper intensity jumps and visible stair steps. Neither is a ground-truth reconstruction: a single photographed surface provides no reference image for PSNR or geometric accuracy.

The near-zero four-corner fit residual only confirms that the computed homography maps its own four constraints correctly. It does not measure corner annotation accuracy, surface planarity, or the unknown physical aspect ratio.

A.4 Blending into a shared frame

Pairwise homographies are composed into the selected reference frame. The reference image keeps the identity transform. All transformed source corners determine a common canvas, and every source is inverse-warped into it using bilinear interpolation.

Feathering

$$M(u,v) = \Sigma_i \, w_i(u,v) \, I_i(u,v) \; / \; \Sigma_i \, w_i(u,v)$$

Each source receives a tent-shaped weight proportional to its distance from the nearest image boundary, normalized to one at its maximum. A positive one-pixel floor preserves isolated edge pixels. The image and its weights are warped together; invalid samples receive zero weight. The overlap is a normalized weighted average, so no image simply overwrites its neighbor.

Feathering reduces hard intensity seams, but it cannot repair a wrong transformation or remove two projections of a moving object. Transparent canvas regions indicate no source coverage. The full projective footprint is retained, making edge stretching and coverage visible.

A.4 +

Automatic seams preserve detail

The indoor scenes additionally use an automatic compositor after geometric estimation. Registered views are ordered by their projected horizontal centers. For each incoming view, a custom dynamic program chooses a top-to-bottom seam through the overlap. It favors similar colors and avoids strong edges, so large misaligned objects are less likely to be cut or averaged.

The seam cost uses RGB absolute disagreement smoothed at σ = 3 pixels, plus 0.6 times smoothed gradient strength and a 0.012 overlap-centering penalty. It is evaluated every four pixels; each step may move up to three sampled columns. A 2-pixel Gaussian transition forms the intermediate composite used to compare the next view.

Final blending decomposes each source into full-resolution Gaussian/residual bands at σ = 3, 12, and 48 pixels. Ownership masks use σ = 1, 3, 12, and 48 pixels from finest to coarsest band. Weights are normalized over valid sources. Fine detail changes narrowly at the seam; wider low-frequency transitions soften exposure differences. The same settings are used for both indoor scenes and both alignment methods.

Minimum-cost seam

$$D(y,x) = C(y,x) + \min_{(|\delta|\le 3)} [D(y-1,x+\delta) + 0.002|\delta|]$$

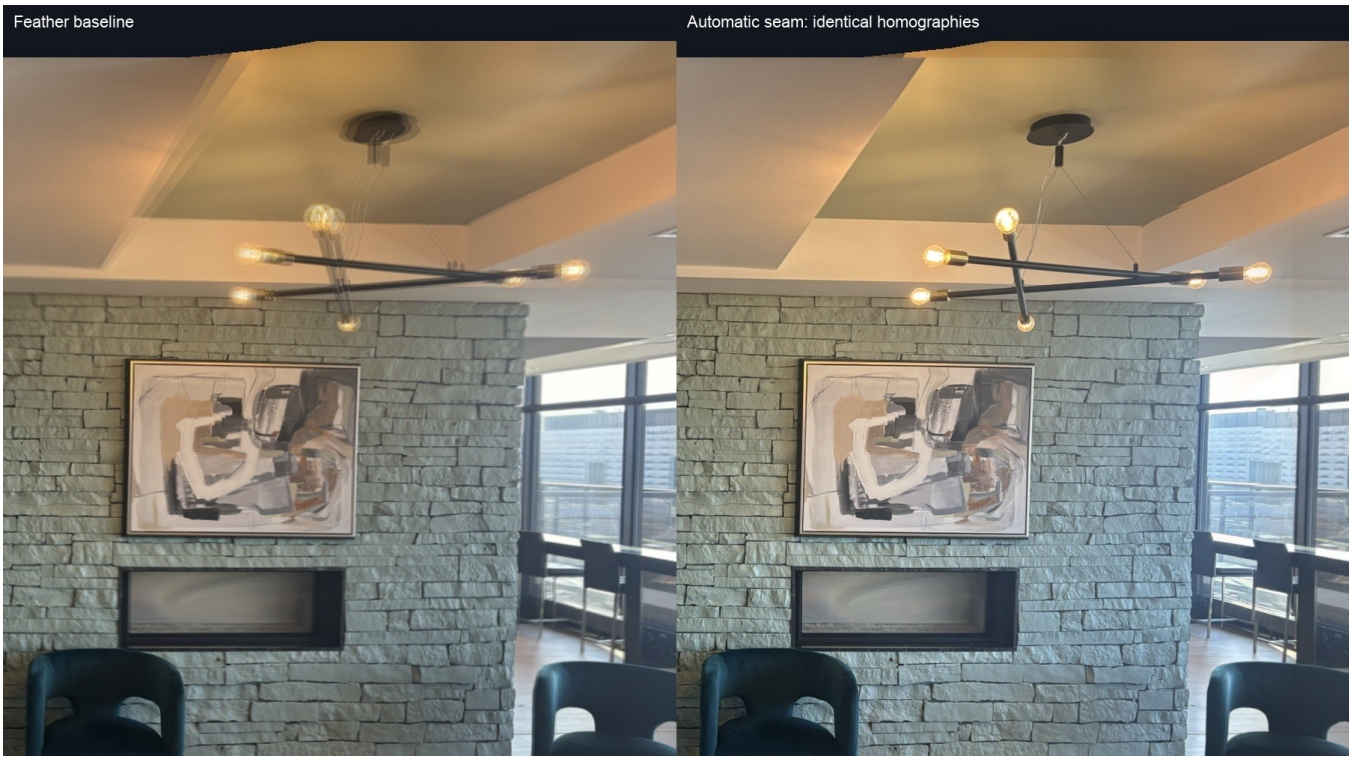


Figure 23. Lounge with identical automatic homographies: feather baseline (left), automatic seams and multiband blending (right). The chandelier and masonry are clearer. Source coverage is unchanged; local seams and parallax can still remain. Sequential vertical seams assume a predominantly horizontal panorama.

A.4 / 01

Basketball court



Figure 24. Basketball court: manually estimated alignment of 3 source images. Reference: IMG_1243. Canvas: 3072 × 1588 pixels.

Twelve manually observed window-frame landmarks constrain each adjacent pair. After manual alignment, automatic image-content seams and multiband blending keep the hoop intact. The feather-only baselines remain available for comparison.

Measured shared-canvas warping and blending: 2.405 s. This is the single-run feather-baseline stage timing and excludes image loading, manual annotation, pairwise fitting, and optional blend refinements.

A.4 +

An intact hoop from automatic seams

The initial feathered mosaic doubled the backboard and rim because the foreground object occupies different locations across the views. The automatic seam refinement keeps it intact while retaining the estimated homographies.

No hoop detector or manually specified rectangle is used. The same algorithm and parameters are applied to manual and automatic geometry in both indoor scenes. The coverage masks are verified to equal those of the feather baselines. This changes source selection, not the geometric estimate.



Figure 25. Automatic matching and homography estimation with the original feather blend, before the compositing refinement.



Figure 26. Identical automatic homographies before and after compositing. Left: feathering averages displaced edges. Right: a seam chosen from image disagreement and edge strength avoids the hoop, retaining a coherent backboard, rim, and net. All content comes from the supplied photographs.

Unmodified feather baselines: Manual · Automatic · Both retain the same homographies as the displayed refinements.

A.4 / 02

Lounge



Figure 27. Lounge: manually estimated alignment of 4 source images. Reference: IMG_1248. Canvas: 2873 × 1583 pixels.

Nine, ten, and ten visually selected landmarks constrain the adjacent pairs. Reviewing the original photographs resolved ambiguous frame corners and wall junctions, lowering fitting RMS errors from 3.04/3.21/2.83 to 1.46/0.95/1.04 working pixels. Automatic seams then preserve clearer masonry, artwork, and chandelier detail. These are fit residuals, not independent accuracy measurements.

Measured shared-canvas warping and blending: 2.822 s. This is the single-run feather-baseline stage timing and excludes image loading, manual annotation, pairwise fitting, and optional blend refinements.

A.4 / 03

City panorama



Figure 28. City panorama: manually estimated alignment of 3 source images. Reference: IMG_1252. Canvas: 3048 × 1687 pixels.

Ten and eleven visually selected landmarks constrain the two adjacent pairs. Refining ambiguous roof tips and using clear roof corners reduced the fitting residuals; the rooflines and chimney now remain sharp across most overlaps.

Measured shared-canvas warping and blending: 2.504 s. This is the single-run feather-baseline stage timing and excludes image loading, manual annotation, pairwise fitting, and optional blend refinements.

B.1

Harris corners and spatial coverage

The supplied Harris implementation detects single-scale corner candidates in a grayscale image. Its 20-pixel border exclusion leaves room for descriptor windows. Harris responses alone can cluster many detections around a textured object; adaptive non-maximal suppression spreads the selected features across the available image structure.

ANMS radius $r_i = \min_j \|x_i - x_j\|_2$, subject to $f_i < 0.9\, f_j$

For each candidate i, the radius is its distance to the nearest sufficiently stronger corner. The largest radii are retained, up to **500 features per image**. Candidates without a stronger suppressor receive a finite radius larger than the candidate cloud's diagonal; ties are resolved by response and input index. The implementation evaluates all candidate pairs in bounded-memory chunks. Descriptor validation can subsequently remove low-contrast patches.

Reference scene	Image	All Harris candidates	Retained descriptors
Basketball court	IMG_1243	30,747	500
Lounge	IMG_1248	38,659	500
City panorama	IMG_1252	37,224	500

All ten source images retain 500 valid descriptors. The figures below show each reference image at full extent and the same enlarged region before and after ANMS. Every detected candidate is plotted.

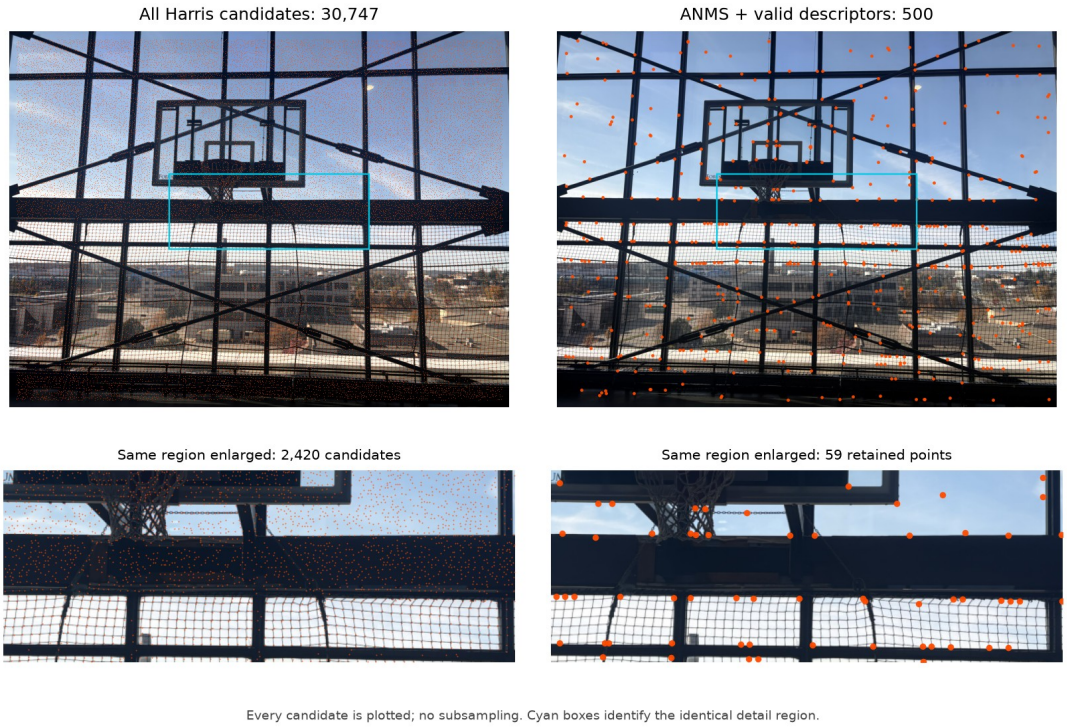


Figure 29. Basketball court, IMG_1243: all Harris candidates on the left; selected ANMS features with valid descriptors on the right. Tiny translucent markers preserve the photograph beneath every candidate; the lower row enlarges the same cyan-box region in both columns. No points are subsampled.

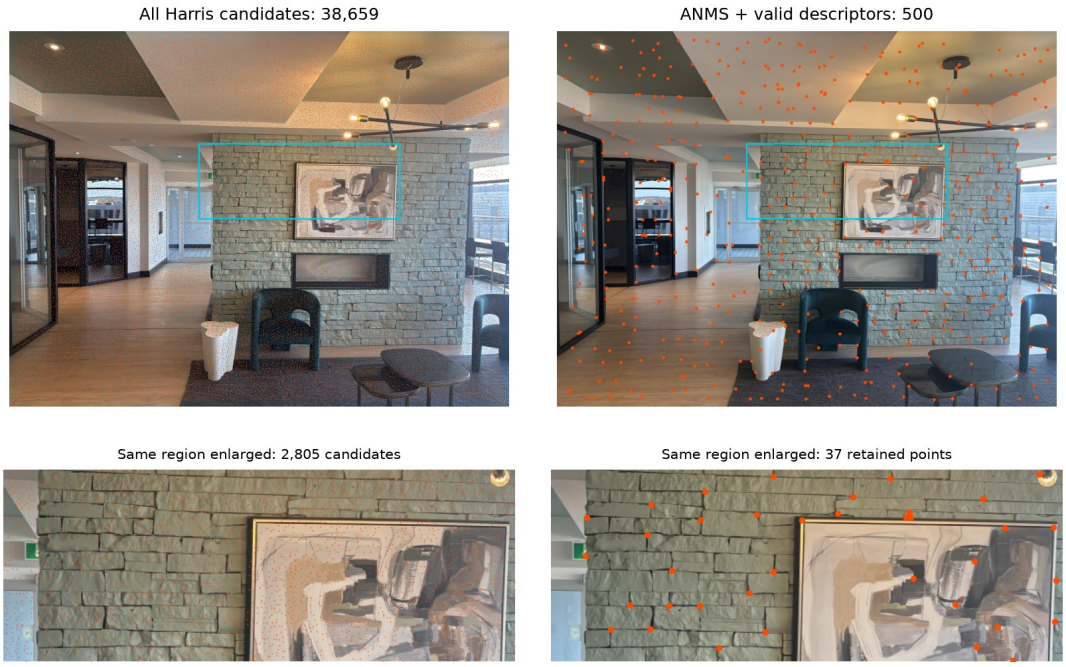


Figure 30. Lounge, IMG_1248: all Harris candidates on the left; selected ANMS features with valid descriptors on the right. Tiny translucent markers preserve the photograph beneath every candidate; the lower row enlarges the same cyan-box region in both columns. No points are subsampled.

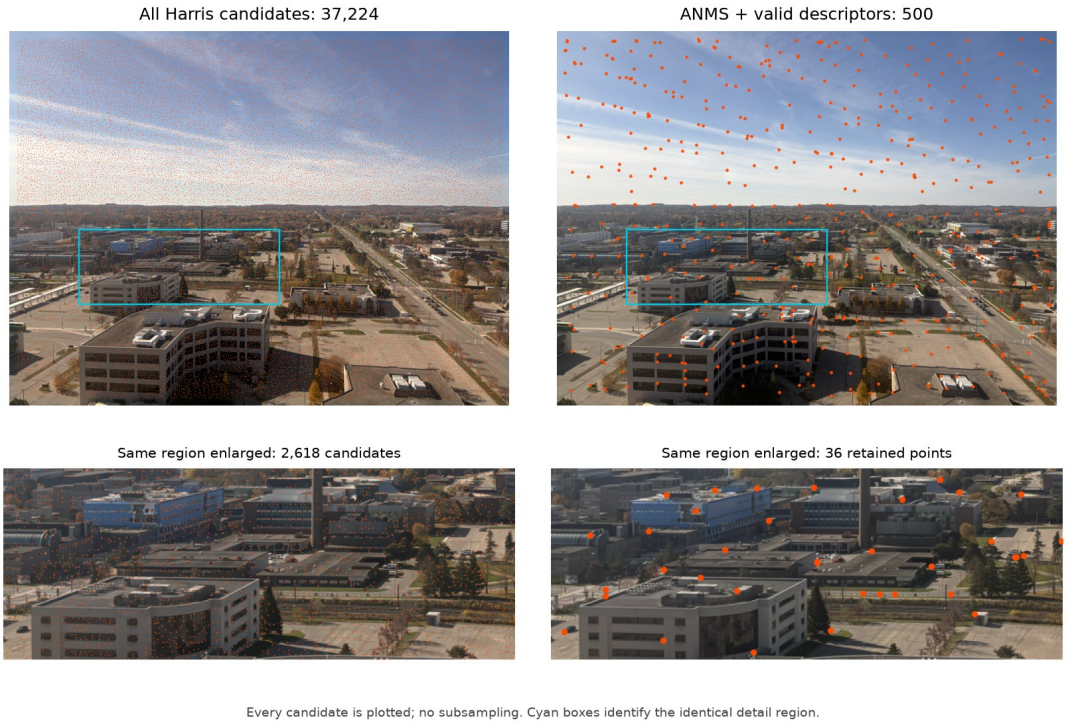


Figure 31. City panorama, IMG_1252: all Harris candidates on the left; selected ANMS features with valid descriptors on the right. Tiny translucent markers preserve the photograph beneath every candidate; the lower row enlarges the same cyan-box region in both columns. No points are subsampled.

B.2

A descriptor from a larger neighborhood

A small patch is more useful when it summarizes a larger neighborhood. Each descriptor is an axis-aligned 8 × 8 grid sampled from a **40 × 40-pixel window**, after Gaussian smoothing with σ = 2 pixels. Grid samples are five pixels apart, at the centers of the eight bins. The sampling uses the same explicit four-neighbor bilinear formula, without a library interpolation call.

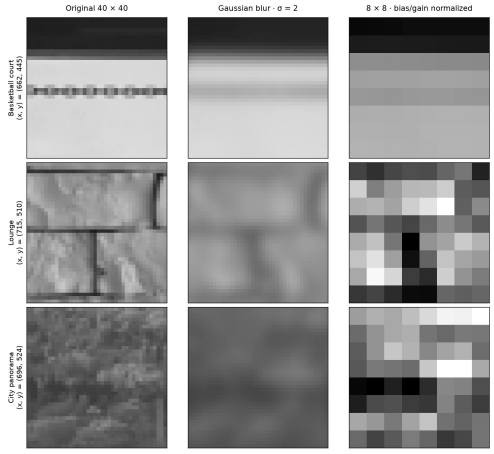


Figure 32. One actual matched feature from each reference image. Left: the original 40 × 40 grayscale neighborhood. Middle: the same neighborhood after Gaussian blur (σ = 2). Right: eight samples per axis, five pixels apart, normalized to zero mean and unit standard deviation. Normalized displays share the range −2 to +2; the exact 64 values are saved with the figure data.

Bias / gain normalization

$$d = (p - \mathrm{mean}(p)) / \mathrm{std}(p), \quad p \in \mathbb{R}^{64}$$

Mean subtraction removes an additive intensity offset; division by the standard deviation removes positive multiplicative gain. Nearly constant patches are discarded instead of dividing by a negligible standard deviation. This normalization does not make descriptors rotation-invariant or scale-invariant. The implementation follows the assignment's single-scale, axis-aligned simplification of MOPS.

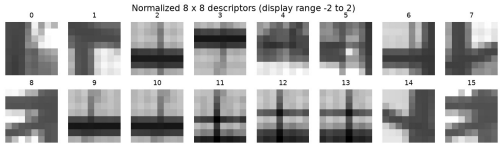


Figure 33. Basketball court: several normalized 8 × 8 descriptors from IMG_1243. Display grayscale is clipped to −2 through +2 for a consistent visual scale.

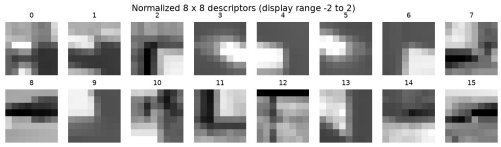


Figure 34. Lounge: several normalized 8 × 8 descriptors from IMG_1248. Display grayscale is clipped to −2 through +2 for a consistent visual scale.

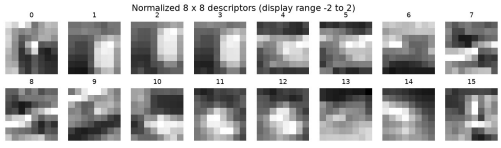


Figure 35. City panorama: several normalized 8 × 8 descriptors from IMG_1252. Display grayscale is clipped to −2 through +2 for a consistent visual scale.

B.3

Matching distinctive features

For every source descriptor, compute squared Euclidean distance to all target descriptors and find the nearest and second-nearest neighbors. A match must be clearly better than its closest alternative. This rejects many repeated or otherwise ambiguous patches before geometric fitting.

Ratio test $\rho = \|d_{source} - d_{nearest}\|_2^2 \,/\, \|d_{source} - d_{second}\|_2^2 < 0.5$

The configured threshold **0.5 applies to squared distances**, equivalent to 0.707 on ordinary Euclidean distances. Equal-distance ties and ambiguous 0/0 matches are rejected. The photographic pipeline uses the forward ratio test; reciprocal nearest-neighbor filtering is available in code but is not enabled for these results.

The ratio test measures descriptor distinctiveness, not geometric correctness. Green and red lines in the following figures show the later RANSAC decision: green matches support the estimated projective model, and red matches were rejected by its reprojection threshold.

Scene	Image pair	Ratio-test matches	Median accepted ratio
Basketball court	IMG_1242 → IMG_1243	75	0.280
Basketball court	IMG_1243 → IMG_1244	96	0.322
Lounge	IMG_1247 → IMG_1248	173	0.159
Lounge	IMG_1248 → IMG_1250	191	0.133
Lounge	IMG_1250 → IMG_1249	257	0.141
City panorama	IMG_1251 → IMG_1252	153	0.205
City panorama	IMG_1252 → IMG_1253	127	0.247

B.3

Correspondences · Basketball court

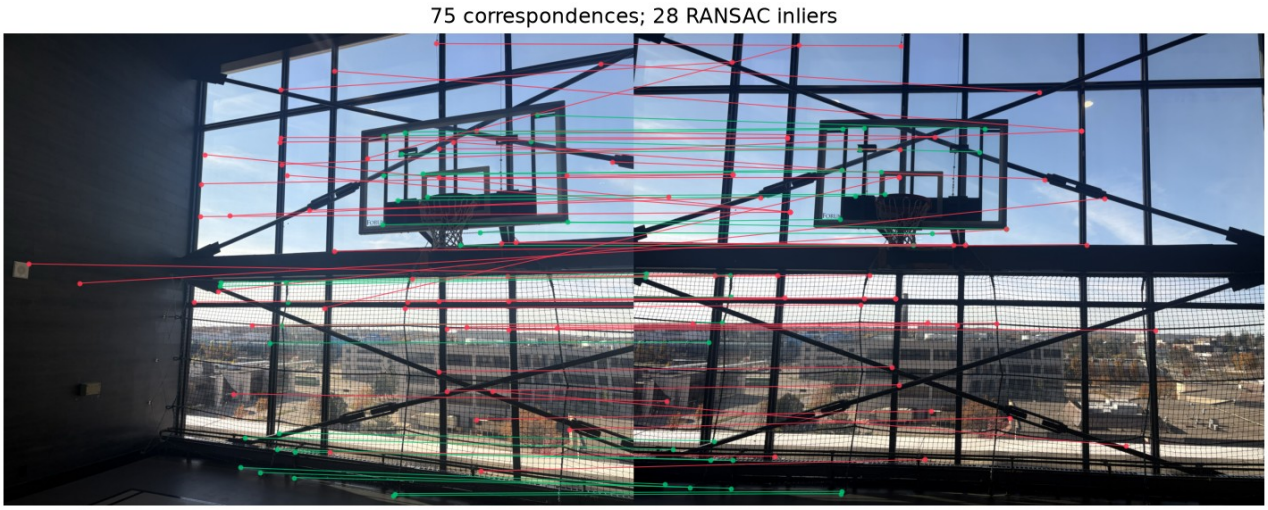


Figure 36. IMG_1242 → IMG_1243: 75 ratio-test matches; 28 RANSAC inliers (green) and 47 rejected matches (red).

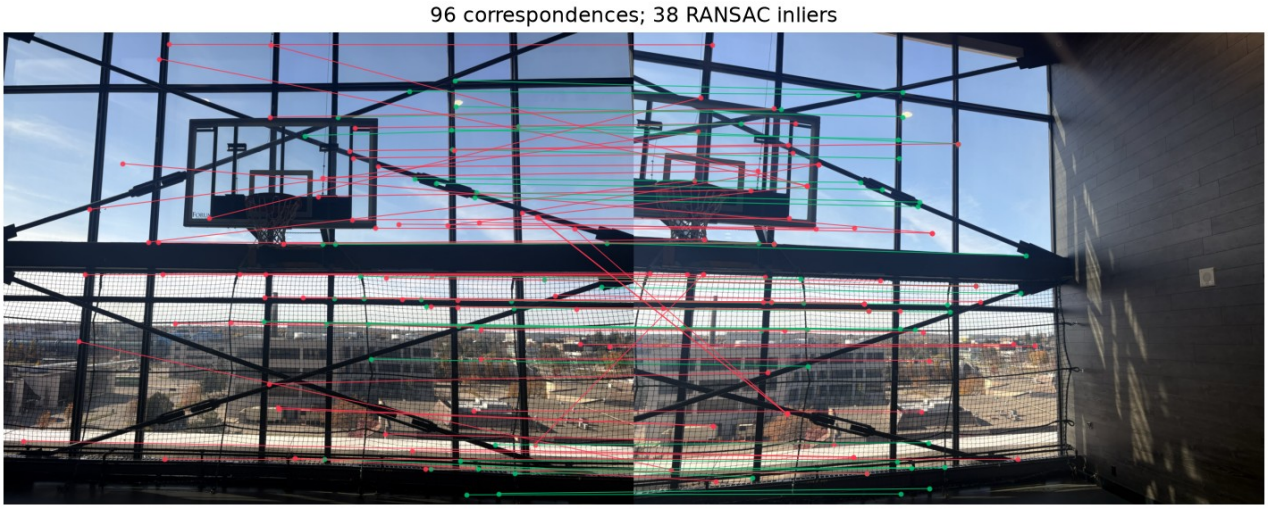


Figure 37. IMG_1243 → IMG_1244: 96 ratio-test matches; 38 RANSAC inliers (green) and 58 rejected matches (red).

B.3

Correspondences · Lounge

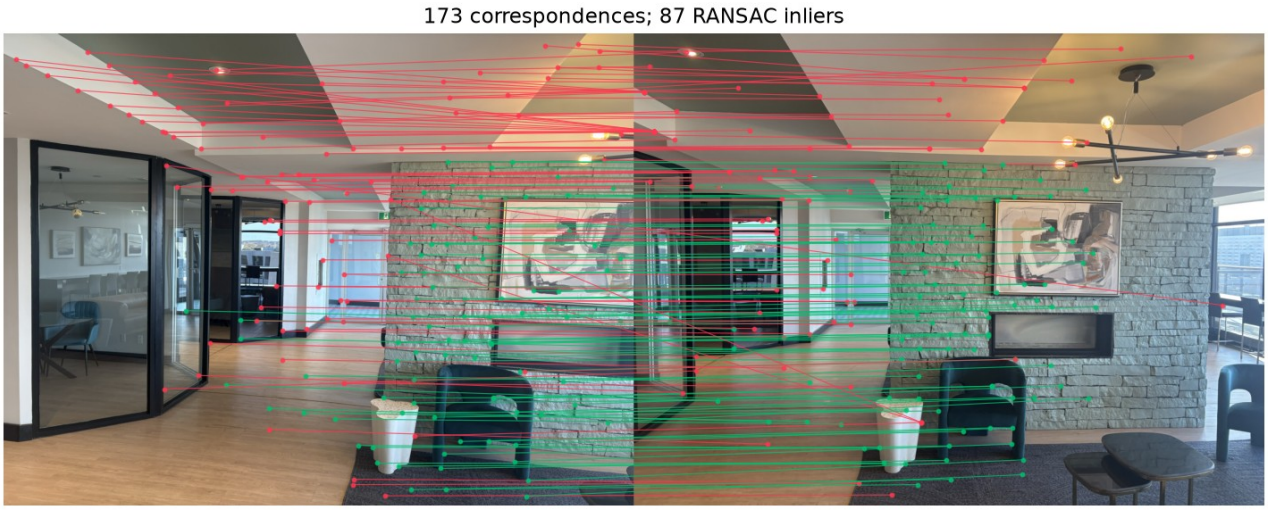


Figure 38. IMG_1247 → IMG_1248: 173 ratio-test matches; 87 RANSAC inliers (green) and 86 rejected matches (red).

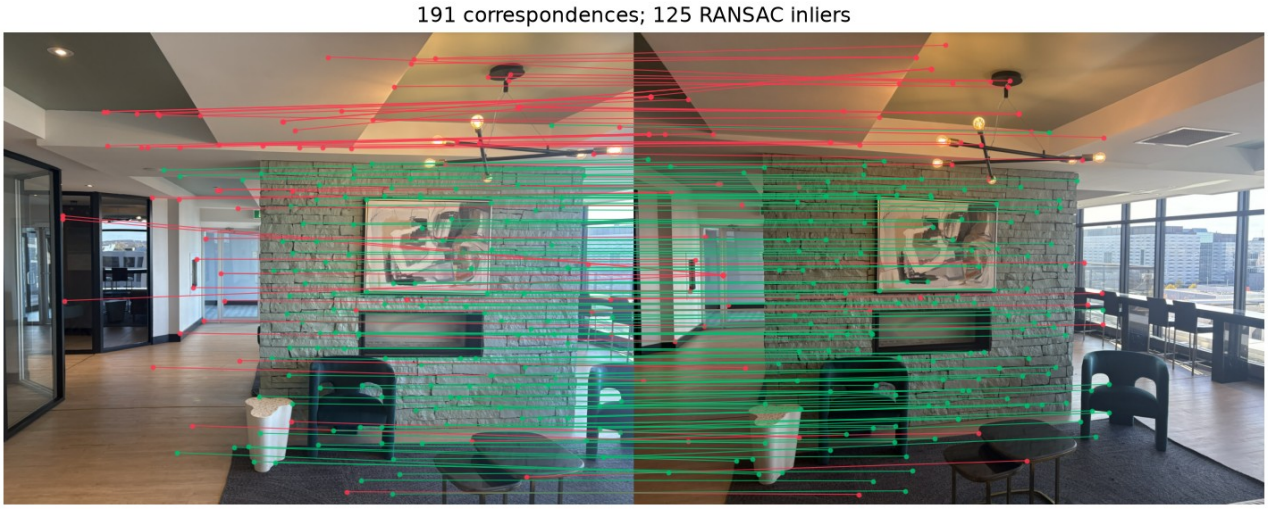


Figure 39. IMG_1248 → IMG_1250: 191 ratio-test matches; 125 RANSAC inliers (green) and 66 rejected matches (red).

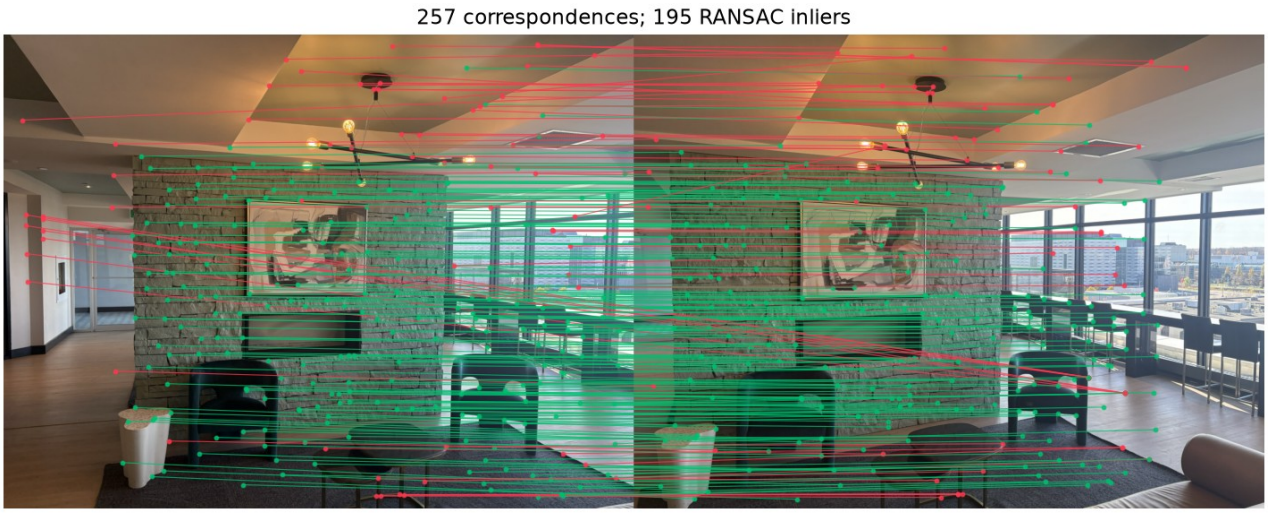


Figure 40. IMG_1250 → IMG_1249: 257 ratio-test matches; 195 RANSAC inliers (green) and 62 rejected matches (red).

B.3

Correspondences · City panorama

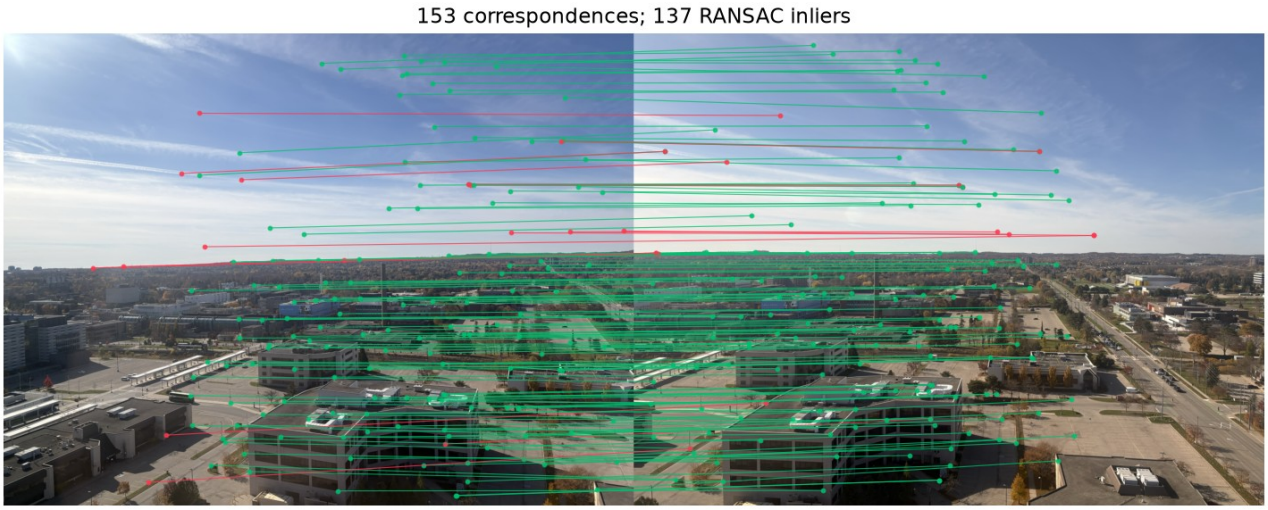


Figure 41. IMG_1251 → IMG_1252: 153 ratio-test matches; 137 RANSAC inliers (green) and 16 rejected matches (red).

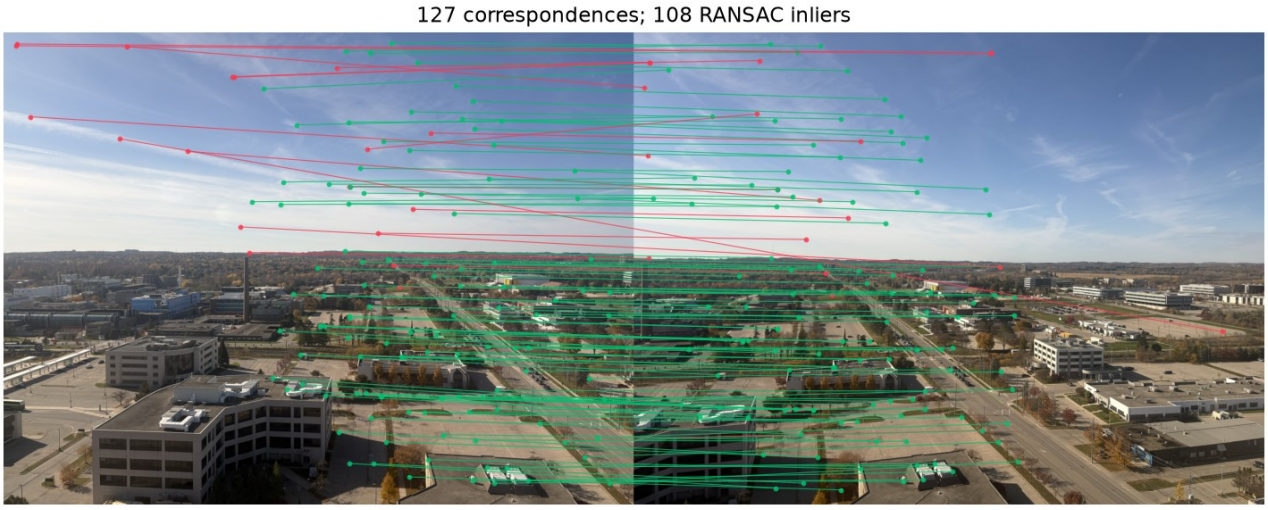


Figure 42. IMG_1252 → IMG_1253: 127 ratio-test matches; 108 RANSAC inliers (green) and 19 rejected matches (red).

B.4

A robust homography from noisy matches

Descriptor matches include outliers. A custom RANSAC loop repeatedly samples four distinct correspondences, fits a homography with `computeH`, and scores every match using forward reprojection distance. Degenerate samples are skipped. Hypotheses are ranked first by inlier count and then by median inlier error.

Inlier test

$$e_i = \|\text{project}(H\,[x_i, y_i, 1]^T) - [u_i, v_i]^T\|_2 \le 3 \text{ pixels}$$

The trial budget adapts to the observed inlier fraction using four-point sampling and a target confidence of 0.995, with a maximum of 5,000 attempts. The best consensus is refitted using all its inliers, with repeated refinement while support remains stable or improves. The final mask is recomputed from the returned homography.

Adaptive trial estimate

$$N = \text{ceil}\left(\frac{\log(1 - \text{confidence})}{\log(1 - \text{inlier_fraction}^4)}\right)$$

The run uses seed 671 plus the pair index. A scene pair must have at least 8 inliers and an inlier fraction of at least 20%. These are minimum support checks, not guarantees of a correct photograph-wide alignment.

Scene	Pair	Inliers / matches	Support	Trials	Median inlier error (px)	Match + fit (s)
Basketball court	IMG_1242 → IMG_1243	28 / 75	37.3%	271	1.125	0.059
Basketball court	IMG_1243 → IMG_1244	38 / 96	39.6%	403	1.224	0.120
Lounge	IMG_1247 → IMG_1248	87 / 173	50.3%	146	1.012	0.047
Lounge	IMG_1248 → IMG_1250	125 / 191	65.4%	81	1.186	0.041
Lounge	IMG_1250 → IMG_1249	195 / 257	75.9%	21	1.053	0.014
City panorama	IMG_1251 → IMG_1252	137 / 153	89.5%	18	1.096	0.014
City panorama	IMG_1252 → IMG_1253	108 / 127	85.0%	23	1.227	0.014

Residuals are forward errors on the fitted consensus, not held-out accuracy. Pair timing includes descriptor matching, RANSAC, and fitting; it excludes feature detection, image loading, and output generation. Trial counts include rejected degenerate samples.

B.4 / 01

Manual vs automatic · Basketball court

The source photographs, working resolution, reference image, bilinear sampler, and automatic seam selection with multiband blending are shared. The correspondence selection and resulting homographies differ.



Figure 43. Manual alignment · Basketball court. Full projective canvas: 3072 × 1588 pixels.



Figure 44. Automatic alignment · Basketball court. Full projective canvas: 3055 × 1557 pixels.

Both results use the same automatic seam algorithm and multiband settings. The backboard, rim, and net remain intact without any hand-selected source region. Their sharpness demonstrates improved compositing; the lower-window crop shows that local geometric disagreements still remain.



Figure 45. Manual alignment (left) and automatic alignment (right), both with automatic seams: lower-window detail cropped at identical reference coordinates.

What remains imperfect

The backboard, net, window structure, and distant exterior lie at different depths. A single homography cannot align all of them if the camera translates. Automatic source selection reduces duplicated edges but does not correct parallax; a seam can still cross another object.

B.4 / 02

Manual vs automatic · Lounge

The source photographs, working resolution, reference image, bilinear sampler, and automatic seam selection with multiband blending are shared. The correspondence selection and resulting homographies differ.



Figure 46. Manual alignment · Lounge. Full projective canvas: 2873 × 1583 pixels.



Figure 47. Automatic alignment · Lounge. Full projective canvas: 2771 × 1559 pixels.

Both results now use the same automatic seam and multiband compositor. Refined manual landmarks produce a clearer fireplace wall than the initial manual fit, while the automatic result also maintains coherent stone courses and artwork edges. Narrow detail transitions avoid broad duplicated chandelier arms and bulbs.



Figure 48. Manual and automatic estimates inspected at corresponding scene details.

What remains imperfect

The furniture, chandelier, corridor, and fireplace wall occupy different depths. Source selection reduces their visible doubling, but it cannot make a single homography explain all viewpoints. Local discontinuities and exposure transitions can remain, particularly near furniture and the ceiling.

B.4 / 03

Manual vs automatic · City panorama

The source photographs, working resolution, reference image, bilinear sampler, and the feathered compositor are shared. The correspondence selection and resulting homographies differ.



Figure 49. Manual alignment · City panorama. Full projective canvas: 3048 × 1687 pixels.



Figure 50. Automatic alignment · City panorama. Full projective canvas: 3012 × 1659 pixels.

Both methods recover a coherent panorama from the same three photographs. The manual fits use 10 and 11 carefully observed landmarks, while the automatic fits retain 137/153 and 108/127 descriptor matches. These are fitting statistics, not independent accuracy measurements.

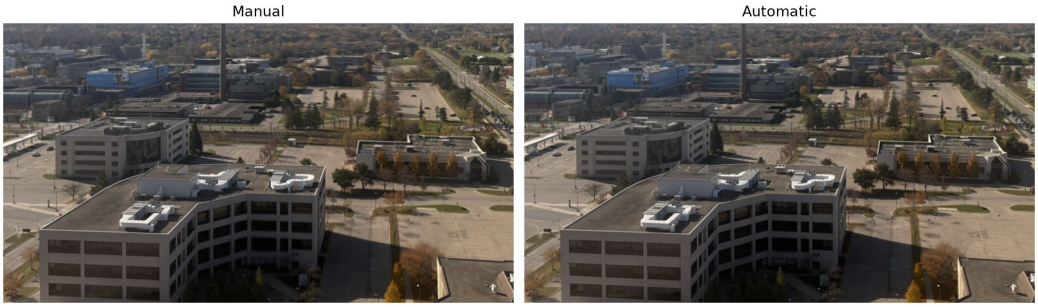


Figure 51. Manual and automatic estimates inspected at corresponding scene details.

What remains imperfect

Most landmarks are below the horizon, so the upper sky and outer image boundaries are extrapolated. Thin roof edges remain sensitive to small point-placement errors; the clouds are not assumed to be fixed geometric landmarks.

09

Validation and limitations

The photographic results test whether the components work together on real data, while known-answer tests check implementation behavior. A synthetic diagnostic is kept separate from the three original-photo mosaics and is not counted as a photographic submission example.

59 automated tests passed. Checks cover homography recovery and degeneracy, inverse sampling and valid black pixels, ANMS and descriptors, ratio matching, RANSAC with outliers, transform chains, and project configuration. Ten seam-specific tests include an exhaustive minimum-path comparison, a shifted-object case, source validity, coverage, constant-field reconstruction, and band blending. Earlier compositor tests remain as regression checks.

```
PYTHONPATH=src .venv/bin/python -m unittest discover -s tests -v
```

Known synthetic check	Measured result
Expected horizontal transform	-160 px
Matches / RANSAC inliers	260 / 250
Held-out maximum reprojection error	1.42e-13 px
Maximum mosaic intensity error	2.07e-13

This generated test has an exactly known translation and noiseless overlap. Its small numerical errors validate that controlled case; they do not predict the accuracy of handheld photographs.

Implementation boundary

The homography system, inverse mapping, nearest and bilinear sampling, feathering, seam dynamic program, multiband reconstruction, ANMS, descriptor normalization, ratio matching, and RANSAC are implemented explicitly. NumPy provides linear algebra; SciPy provides Gaussian smoothing; the supplied Harris implementation is used as permitted. Pillow handles image I/O and orientation, and matplotlib produces diagnostic figures.

Limitations that affect interpretation

- A global homography cannot explain parallax, motion, or lens distortion across an arbitrary handheld scene.
- Axis-aligned, single-scale patches can fail under large rotation, scale changes, low texture, or repeated structures.
- Feathering averages disagreements; it does not remove geometric double edges or exposure differences.
- Consensus residuals and four-corner fitting residuals are not independent accuracy measurements.
- The rectified books use approximate display proportions rather than measured physical dimensions.

10

Reproducibility and references

The same configuration drives both manual and automatic processing. Input provenance, original-to-working transforms, correspondences, recovered matrices, RANSAC decisions, canvas bounds, and measured times are retained as JSON. The report builder reads those records directly.

Parameter	Value
Working longest side	1400 px
Maximum retained features	500
ANMS robust strength factor	0.9
Descriptor window / grid	40 × 40 / 8 × 8; 5-pixel sample spacing
Descriptor Gaussian σ	2 pixels
Squared-distance ratio threshold	0.5
RANSAC forward-error threshold	3.0 working-image pixels
RANSAC seed	671 + pair index
Baseline interpolation / blending	Bilinear / distance-to-boundary feathering
Indoor refinement	Automatic content seams; Gaussian bands σ = 3, 12, 48 px

Reproduce the experiment

```
python3 -m venv .venv
.venv/bin/pip install -r requirements-lock.txt
PYTHONPATH=src .venv/bin/python -m unittest discover -s tests -v
.venv/bin/python run_project.py --mode all
.venv/bin/python refresh_corner_figures.py
.venv/bin/python run_rectifications.py
.venv/bin/python run_seam_refinement.py
.venv/bin/python make_report_assets.py
.venv/bin/python run_synthetic.py
.venv/bin/python build_report.py
```

Run from the assignment folder. The input archive and configuration preserve the original-photo annotations; figure and report generation use the recorded results. Hardware and concurrent activity can change timings.

References

1. SYDE 671, Assignment 3: *[Auto]Stitching Photo Mosaics*. Course handout; adapted from Berkeley CS180 Assignment 3.
2. Brown, M., Szeliski, R., and Winder, S. (2005). *Multi-Image Matching using Multi-Scale Oriented Patches*. CVPR. The implementation follows the assignment's simplified single-scale, axis-aligned variant.
3. Assignment-supplied `harris.py`: Harris candidate detection and distance helper.

APX

Automatic homography matrices

These full-image pairwise transformations map source working coordinates into target working coordinates. The JSON download retains full precision and all image-to-reference compositions.

Basketball court · IMG_1242 → IMG_1243

```
1.66912  −0.0481914  −743.124
0.302618   1.44275  −285.728
0.000480189   9.73834e−06   1
```

Basketball court · IMG_1243 → IMG_1244

```
1.75943  −0.0620795  −835.532
0.332926   1.47492  −286.537
0.000542659  −2.53564e−05   1
```

Lounge · IMG_1247 → IMG_1248

```
1.39974   0.0190938  −492.918
0.11684   1.22941  −92.1788
0.000302281  −1.80218e−05   1
```

Lounge · IMG_1248 → IMG_1250

```
1.37391   0.00845299  −452.899
0.126847   1.22117  −110.258
0.000262928  −2.65058e−08   1
```

Lounge · IMG_1250 → IMG_1249

```
1.26089   0.00617285  −278.296
0.0834213   1.14026  −59.5377
0.000183846  −6.76548e−06   1
```

City panorama · IMG_1251 → IMG_1252

```
1.5937  −0.0133569  −607.481
0.244937   1.37944  −231.215
0.000415361   2.38386e−05   1
```

City panorama · IMG_1252 → IMG_1253

```
1.8885  −0.00571289  −981.234
0.336389   1.5636  −279.178
0.000637776  −3.16932e−05   1
```